BRAVO Challenge Report - Ensembles

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Abstract Participating in the BRAVO challenge, which focuses on semantic segmentation robustness, we investigate the applicability of ensembles to this domain. We focus on two potential issues that need to be addressed. First, we evaluate ways of increasing the sources of diversity in an ensemble with respect to datasets and models by aggregating predictions generated from different network architectures and trained on different datasets. Second, we examine the impact of network architecture on ensemble performance by using modern networks known to perform well. We obtain competitive results on the BRAVO challenge, demonstrating the utility of diverse models in our ensemble as well as how critical the choice of architecture can be on performance. Our experiments show that ensembles may still be viable to address semantic segmentation robustness, in contrast to their low ranking in some comparative leaderboards.

1 Introduction and Motivation

While the task of segmentation has been a staple of computer vision nearly since its inception [10], the more modern task of semantic segmentation has also been studied for many years [17] with its own set of challenging datasets [6]. The goal is to assign a label to each pixel in an image representing what category the object belongs to; typical examples include cars, roads and buildings.

One of the potential issues with applying a semantic segmentation algorithm in the real world comes from *out of distribution* objects, where the algorithm is presented with objects which it has never seen before during training and thus may behave in unexpected ways, such as by being very confident in asserting that said unknown object (*e.g*. an animal) is in fact something it is not (*e.g*. a car) [2]. On top of this, non-ideal environmental conditions such as direct sunlight on the camera, fog or rain can hamper the detection of known objects by obscuring them or changing their appearance to the algorithm in some way [18].

Thus, when deploying a semantic segmentation algorithm in the real world to address a particular task, such as autonomous driving, it is essential to be able to both correctly identify known classes seen during training even under adverse conditions as well as identify *out of distribution* objects. This task is generally referred to as achieving robustness, and is the main motivation behind the BRAVO challenge where the evaluation requires both correctly identifying known classes under adverse conditions as well as recognizing out-of-distribution objects. There are a number of existing algorithms that can help address the task, including the use of ensembles or ensemble-like methods such as Deep Ensembles [8] and MC Dropout [7].

Ensembles are usually generated from variations in model parameters, with the mean across models (sometimes referred to as samples) serving as the aggregation operator allowing for a prediction to be made. Meanwhile, the variance across models (including through statistical measures such as Shannon entropy [16]) can serve to identify uncertain predictions. However, examining the leaderboards of challenges such as SMIYC (Segment Me If You Can) [3] and past work on out of distribution detection [1, 13] shows that ensemble-based methods have serious limitations, with more recent approaches declining to even compare against them.

Given that the theory of ensembles remains unchanged, we question this poor performance in the domain of robust semantic segmentation. We hypothesize that ensembles are encumbered in two ways and investigate our beliefs on the BRAVO challenge. First, we note that popular methods such as MC Dropout or Deep Ensembles relying on variations in the weights of a fixed network architecture to produce several models. In practice, we have observed such variations to be minimal, and as such we aim to test if greater model diversity (*i.e.* more variation between models) can lead to improved performance. Second, we note that approaches such as MC Dropout or Deep Ensembles in practice are only ever evaluated with a single architecture and inherit its performance characteristics. Thus, the choice of architecture on which to use an ensemble sets the baseline performance to be expected. With this in mind, we aim to determine if ensembles that use more modern architectures are more competitive against the current state of the art than past attempts.

2 Approach

2.1 Theory

The solutions to both tracks involves ensembles, albeit in different configurations. For both of them, we use ensembles in a standard configuration where we have Q models: [11, 12]

$$\left\{P(y \mid \boldsymbol{x}^*, \boldsymbol{\theta}^{(q)})\right\}_{q=1}^{Q}, \quad \boldsymbol{\theta}^{(q)} \sim p(\boldsymbol{\theta}|\mathcal{D}). \tag{1}$$

each one of which is capable of generating a prediction y from a test input $\boldsymbol{x}^*$ with weights $\boldsymbol{\theta}^{(q)}$ constrained by the prior $p(\boldsymbol{\theta})$. The predictions of these models can be aggregated through the predictive posterior as the mean across models *i.e.*:

$$P(y \mid \boldsymbol{x}^*, \mathcal{D}) = \frac{1}{Q}\sum_{q=1}^{Q} P(y \mid \boldsymbol{x}^*, \boldsymbol{\theta}^{(q)}). \tag{2}$$

With $P(y \mid \boldsymbol{x}^*, \mathcal{D})$, we now have a confidence assigned to each class, with the maximum across the set of classes providing the predicted class and associated confidence in the prediction as required by the BRAVO challenge.

2.2 Methodology

For Track 1, we address both aspects of our hypothesis. For the first part on model diversity, we chose to use ensembles of two models: Mask2Former [4] and HMSA

(Hierarchical Multi-Scale Attention for Semantic Segmentation) [19]. In all cases for this track pretrained Cityscapes models available online are used. Note that for the Mask2Former approach we build off the code and use models provided by the authors of RbA (Rejected By All) [13], whose main contribution is a scoring function that takes as input the output of a Mask2Former model. We explicitly denote the approaches where we use said scoring function as being RbA, but otherwise refer to the approach as Mask2Former. Ideally, we would have used more models, but this was not possible due to time and resource constraints. The two models were chosen as they have very different architectures, and thus are good candidates for exploring model diversity in terms of architectures. They also satisfy the other part of our hypothesis as they are known to perform very well in terms of semantic segmentation performance and out of distribution performance, on the Cityscapes [5] and SMIYC [3] leaderboards, providing a good starting point in terms of performance.

For Track 2, our primary goal was to evaluate the potential use of different datasets as a source of model diversity for the ensembles. The BRAVO challenge is set up to evaluate only the standard 19 Cityscapes evaluation classes, and Track 2 allows for the use of multiple datasets (Cityscapes [5], BDD100k [22], Mapillary Vistas [14], India Driving Dataset [20], WildDash 2 [23], GTA5 [15] and SHIFT [18]). With all of these datasets placing a clear focus on autonomous driving in some way, they are all formatted to either use the aforementioned 19 Cityscapes classes or use classes similar enough such that they can be mapped to the Cityscapes ones. This presents an opportunity where we can train models on different datasets with different characteristics (including some synthetic ones) while maintaining compatibility with one another. Here, we use the same models as Track 1, with a particular focus on HMSA as we train a model with that architecture for each allowed dataset.

3 Experiments

3.1 Implementation Details

As shown in Tab. 1, we evaluate a number of different ensemble configurations and compare against some baselines.

RbA Baseline The RbA baselines are obtained by executing the code provided by [13] with the Swin-B and Swin-L models they provide in the Model Zoo (under the heading *Cityscapes Inlier Training*) on the BRAVO images. This provides the logits from the model for each image, with the maximum across classes of said logits providing the predicted class for each pixel. The confidence score for each pixel was generated by applying the RbA score function to the same logits, followed by normalizing the result to the 0-1 range. As the BRAVO challenge expects the confidence in predictions, as opposed to the confidence in out of distribution assertions, the result is inverted (*i.e.* $1 - \text{RbA score}$).

HMSA Baseline The HMSA baseline is generated by using the `nimble-chihuahua` model from HMSA as provided by the authors, using the standard parameters as they specify in their codebase including using scales $0.5, 1.0, 2.0$. The maximum softmax value is used to determine the class and associated confidence at each pixel.

Track 1 models Ensemble Configuration A & C[1] involve combining the predictions of Mask2Former [4] and HMSA [19]. Configuration A consists of two models: the Swin-L model of Mask2Former and the `nimble-chihuahua` model from HMSA. Configuration C adds one additional model in the form of the Swin-B model for Mask2Former, for a total of three models. We apply the softmax oeprator to the logits of each model independently, which then gives us $Q = \{2, 3\}$ models that satisfy Eq. (1), allowing Eq. (2) to be applied giving us a prediction and associated confidence in each class for every pixel.

Track 1 "model selection" The "model selection" approach is not really a fair comparison given its implementation, but we devised it as a comparison for Ensemble Configurations A & C. It takes advantage of the BRAVO dataset's structure and metric calculations, applying the best method depending on the dataset split. This is only possible as the BRAVO challenge does metric calculations on a per-split basis, and as such the confidence in each prediction needs only be commensurate within each split. Additionally, with one exception every split is evaluated for either semantic or out-of-distribution performance, not both. Specifically, on the splits `bravo-ACDC`, `bravo-outofcontext`, `bravo-synflare` and `bravo-synrain` we use the `nimble-chihuahua` model from HMSA. For the `bravo-synobjs` split, we use an ensemble following Ensemble Configuration A above. For the `bravo-SMIYC` split, we use the RbA Swin-L model, including the RbA scoring function.

Track 2 training Before we could evaluate any models, we first needed to obtain one trained model per dataset with the HMSA architecture [19]. The models for each were generated as follows:

- **Cityscapes**: We used the author-provided `nimble-chihuahua` model [19].
- **Mapillary**: After converting the dataset to use Cityscapes labels following Sec. 6.1 we trained the model using transfer learning from the `fast-rattlesnake` model [19] as provided by the authors and with the same training configuration as they provide with the code for their Mapillary model, except the number of epochs, which we set to 15 as we only needed to adapt the model to the new class scheme.
- **GTA5**: We first removed some corrupted images and then resized all images and ground truth masks to (1914, 1052). The model was then trained with transfer learning from the `fast-rattlesnake` model as well, with the same training settings as used by [19] for their *cityscapes_sota* training configuration.
- **SHIFT**: We use the dataset author provided label mapping to Cityscapes and train the model with transfer learning from the `outstanding-turtle` model from [19]. The training parameters are the same as with the GTA5 model, except for the learning rate which is set to 0.005.

[1] Naming confusion in the final hours of the challenge led to configuration B being skipped.

- **BDD100K**: This model is trained via transfer learning from the `industrious-chicken` model from [19] with the same settings as the GTA5 model.
- **IDD**: This transfer learning source for this model is the `outstanding-turtle` model. Parameters are the same as the GTA5 model.
- **Wild Dash 2**: The `nimble-chihuahua` model is used for pretraining in this case, with parameters the same as the SHIFT model.

There are also a number of settings which we set when training all models on each dataset but do not explicitly enumerate in the interest of brevity:

- **Splits**: In all cases we used dataset author-provided training splits.
- **Tile size**: The training process uses tiling during training to try and ensure a better class distribution when training for some classes that are more rare. We try and set this such that each image can be decomposed into two tiles as best as possible, depending on the size(s) of images contained in the dataset.
- **Crop size/resizing**: We set the image resizing and cropping to match the image size(s) in the dataset as best as possible so as to minimize data loss.
- **Epochs**: All models are trained for 175 epochs unless otherwise indicated. However, the final model used was the one which achieved the best results on the respective validation set of each dataset during the entire training run and thus may have been trained for fewer epochs.
- **Batch sizes**: Training and validation batch sizes are set as high as possible given the VRAM capacity of the GPUs being used for training.

Track 2 models Multi-dataset ensemble configuration A uses an ensemble of models with the HMSA architecture, one for each of the allowed datasets. All are trained following Sec. 3.1, generating $Q = 7$ models and allowing us to apply Eqs. (1) and (2) in the same manner as Track 1. Multi-dataset ensemble configuration B uses the same models as A, but adds the two (Swin-L an Swin-B) Mask2Former architecture models used in Track 1, for a total of $Q = 9$ models. The final multi-dataset ensemble configuration C has $Q = 4$ models: the two (Swin-L and Swin-B) Mask2Former models from Track 1, and the Cityscapes and Mapillary models as described in Sec. 3.1.

4 Experimental Results

In Tab. 1, we demonstrate the performance of our ensemble approach, under different configurations and against some baselines, on the BRAVO challenge test set. Note that we only include selected work from others here; please refer to the official leaderboard and ECCV paper for more comparisons against other work.

Given these results, we can make a few observations. The first is that relatively straightforward ensembles using the mean achieves very respectable results, doing relatively well in combining the divergent semantic and OOD performance of the Mask2Former and HMSA models. It is not quite able however to achieve the best of both worlds, as demonstrated by the greater performance of the RbA and HMSA baselines as well as the "model selection" version of ensembles. It is clear as well that other approaches,

	Approach	BRAVO Index		
		BRAVO↑	Semantic↑	OOD↑
Track 1	RbA Swin-B Baseline	0.3773	0.2767	0.5925
	RbA Swin-L Baseline	0.3753	0.2523	0.7324
	HMSA Baseline	0.3600	**0.7060**	0.2416
	Ensemble configuration A (ours)	0.5986	0.6725	0.5393
	Ensemble configuration C (ours)	0.6109	0.6428	0.5820
	Ensemble Model selection* (ours)	0.6349	0.6939	0.5852
	DINOv2 [9], ViT-L, 8x8 patch size, linear decoder	**0.7789**	0.6981	**0.8807**
	PixOOD YOLO (="Model Selection")*	0.6779	0.5706	0.8349
	PixOOD w/ ResNet-101 DeepLab	0.6119	0.5866	0.6395
	PixOOD [21]	0.5347	0.4038	0.7909
Track 2	Multi-dataset ensemble configuration A	0.4063	0.6599	0.2935
	Multi-dataset ensemble configuration B (ours)	0.4554	0.6456	0.3517
	Multi-dataset ensemble configuration C (ours)	0.5879	0.6448	0.5402
	InternImage-OOD	**0.6263**	**0.6934**	0.5710
	PixOOD w/ DeepLab Decoder Cityscapes+BDD100K	0.6081	0.5226	**0.7272**

Table 1: Performance of the approaches we evaluated on the BRAVO challenge, alongside selected work from other authors (DINO, PixOOD, InternImage). *: Not necessarily a fair comparison against other approaches.

such as PixOOD [21], suffer from some trade-offs in terms of semantic and OOD performance when making architecture changes. In this particular case, it is safe to say that network architecture can play a very significant role in performance, and the performance of ensembles can be heavily influenced by it. Our second observation is that it is clear model diversity can play an important role as well, as the performance of our approach is entirely due to being able to combine two approaches that specialize in different metrics. However, with only two models at play for Track 1, we cannot make any definitive statements.

For Track 2, we can see that while using several models helps the BRAVO score for Multi-dataset ensemble configuration A over the HMSA baseline, it does so by improving the OOD score at the expense of the semantic score. Multi-dataset configurations B and C, however, show that adding the two Mask2Former models, with their ability to do better at OOD detection, is much more impactful than using more models of the same HMSA architecture trained on different datasets. More generally, neither of the two approaches tested on both Track 1 and 2 (our approach with Ensembles and PixOOD) show any notable improvement from being allowed to use more datasets.

5 Conclusion

Overall, we demonstrate that our hypothesis of ensembles in the field of robust semantic segmentation being held back by architecture choice and lack of diversity holds some merit. With ensembles, we show that using multiple datasets can help improve the performance but using diverse specialized models works best. Even then, however, the mean of said specialized models has its limits and is not quite able to ideally combine the results of both models. Future work is needed to determine more precisely the impact large amounts of model and dataset diversity can have on performance compared to more traditional work such as MC Dropout, as well as to evaluate alternate methods of aggregating and interpreting the ensemble *e.g.* the use of predictive entropy as a measure of uncertainty.

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6 Appendix

6.1 Mapillary mapping

In Tab. 2, we present the mapping we used to convert the Mapillary Vistas [14] dataset to the Cityscapes format.

Mapillary Vistas Category	**Cityscapes Category**
animal-bird	void
animal-ground-animal	void
construction-barrier-curb	void
construction-barrier-fence	fence
construction-barrier-guard-rail	void
construction-barrier-other-barrier	fence
construction-barrier-wall	wall
construction-flat-bike-lane	road
construction-flat-crosswalk-plain	road
construction-flat-curb-cut	sidewalk
construction-flat-parking	road
construction-flat-pedestrian-area	sidewalk
construction-flat-rail-track	void
construction-flat-road	road
construction-flat-service-lane	void
construction-flat-sidewalk	sidewalk
construction-structure-bridge	void
construction-structure-building	building
construction-structure-tunnel	void
human-person	person
human-rider-bicyclist	rider
human-rider-motorcyclist	rider
human-rider-other-rider	rider
marking-crosswalk-zebra	road
marking-general	road
nature-mountain	void
nature-sand	terrain
nature-sky	sky
nature-snow	terrain
nature-terrain	terrain
nature-vegetation	vegetation
nature-water	void
object-banner	void
object-bench	void
object-bike-rack	void
object-billboard	void
object-catch-basin	void

object-cctv-camera	void
object-fire-hydrant	void
object-junction-box	void
object-mailbox	void
object-manhole	void
object-phone-booth	void
object-pothole	void
object-street-light	void
object-support-pole	pole
object-support-traffic-sign-frame	traffic sign
object-support-utility-pole	pole
object-traffic-light	traffic light
object-traffic-sign-back	void
object-traffic-sign-front	traffic sign
object-trash-can	void
object-vehicle-bicycle	bicycle
object-vehicle-boat	void
object-vehicle-bus	bus
object-vehicle-car	car
object-vehicle-caravan	void
object-vehicle-motorcycle	motorcycle
object-vehicle-on-rails	train
object-vehicle-other-vehicle	void
object-vehicle-trailer	void
object-vehicle-truck	truck
object-vehicle-wheeled-slow	void
void-car-mount	void
void-ego-vehicle	void
void-unlabeled	void

Table 2: The conversion used to convert Mapillary Vistas labels to Cityscapes. Inspired from but different from the mapping presented in [2]